

Lecture 02B

Vector Subspaces

Linear Combinations, Span

MH1201 Linear Algebra 2

Any subspace contains a ^{an additive identity} zero vector.

<https://www.wooclap.com/>

HIDDEN

Which of the following is an example of a zero vector?

To obtain one point, you need to choose ALL correct answers.

1. $(0,0,0,0,0)$ → 5-tuple — zero vector for $\mathbb{R}^5$

2. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ → zero matrix — zero vector for $M_{2 \times 2}$

3. z → zero function ^{→ not a number} — zero vector for $F(\mathbb{R}, \mathbb{R})$

4. 0 → zero — zero vector for $\mathbb{R}$

→ a function that maps all numbers to zero.

Examples of Subspaces

→ non

Proposition. $\mathcal{S}_{n \times n}(\mathbb{F})$ (the set of all symmetric $n \times n$ -matrices) is a subspace of $(\mathcal{M}_{n \times n}(\mathbb{F}), \oplus, \otimes)$.

Today

Proposition. $\mathcal{P}_d(\mathbb{F})$ (the set of all polynomials with degree at most d) is a subspace of $(\mathcal{F}(\mathbb{F}, \mathbb{F}), \oplus, \otimes)$.

Today

Proposition. The set of solutions of a homogenous system is a subspace of $(\mathbb{F}^n, \oplus, \otimes)$.

Proposition. $\mathcal{P}_d$ is a subspace of $\mathcal{F}$.

Here, d is a non negative integer

$\overset{\text{curly P}}{\mathcal{P}_d(\mathbb{F})} \triangleq \{ \text{all polys of deg} \leq d \}$

$$= \{ a_0 + a_1x + a_2x^2 + \dots + a_dx^d : a_0, a_1, \dots, a_d \in \mathbb{R} \}$$

$\overset{\text{degree 1}}{x+x^2 \in \mathcal{P}_1(\mathbb{R})}?$
 $\rightarrow \text{No.}$

$\overset{\text{degree 2}}{x+x^2 \in \mathcal{P}_2(\mathbb{R})}?$
 $\rightarrow \text{yes.}$

$\overset{\text{degree 3}}{x+x^2 \in \mathcal{P}_3(\mathbb{R})}?$
 $\rightarrow \text{Yes}$

$\overset{\text{degree 4}}{x+x^2 \in \mathcal{P}_4(\mathbb{R})}?$
 $\rightarrow \text{Yes}$

Proposition. $\mathcal{P}_d$ is a subspace of $\mathcal{F}$.

$\mathcal{P}_d$ is the set of all polynomials of degree at most d .

Theorem. (Conditions for Subspace)

Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** $\mathbf{0}$ belongs to $\mathcal{U}$.
- **(Closure under Vector Addition).**

For all $u, v \in \mathcal{U}$, we have $u \oplus v \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have $\lambda \otimes u \in \mathcal{U}$.

Recall: $\mathbf{0} \in \mathcal{P}_d(x)$? *zero poly*

Here, we are asking $Z(x)$ belongs to $\mathcal{P}_d$

① Think: $0 + 0x + 0x^2 + \dots + 0x^d \in \mathcal{P}_d$

② What is deg of $Z(x)$?

Convention: deg $Z(x)$ is $-\infty$. *just accept value lol*

So $\deg Z(x) \leq d \Rightarrow Z(x) \in \mathcal{P}_d$

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For $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have $\lambda \otimes u \in \mathcal{U}$.

WTS: $f+g \in \mathcal{P}_d$

Let $f(x) = a_0 + a_1x + \dots + a_dx^d \in \mathcal{P}_d$

$g(x) = b_0 + b_1x + \dots + b_dx^d \in \mathcal{P}_d$

$(f \oplus g)(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_d + b_d)x^d$

$\Rightarrow \deg(f \oplus g) \leq d$

So, $f \oplus g \in \mathcal{P}_d$

↓
this coefficient
may be zero

↓
even so, $\deg(f \oplus g)$ is
at most d

Proposition. $\mathcal{P}_d$ is a subspace of $\mathcal{F}$.

$\mathcal{P}_d$ is the set of all polynomials of degree at most d .

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For all $u, v \in \mathcal{U}$, we have $u \oplus v \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have $\lambda \otimes u \in \mathcal{U}$.

$$\text{WTS } (\lambda \otimes f) \in \mathcal{P}_d$$

$$\text{let } \lambda \in \mathbb{F}$$

$$\text{let } f(x) = a_0 + a_1x + \dots + a_dx^d \in \mathcal{P}_d$$

$$(\lambda \otimes f)(x) = \lambda a_0 + \lambda a_1x + \dots + \lambda a_dx^d \in \mathcal{P}_d$$

$$\deg(\lambda \otimes f) \leq d$$

$$\therefore (\lambda \otimes f) \in \mathcal{P}_d$$

All 3 conditions met,

$\therefore \mathcal{P}_d$ is a subspace

Proposition. The set of solutions of a homogenous system is a subspace of $(\mathbb{F}^n, \oplus, \otimes)$. LA1

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0$$

$\vdots$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0$$

· n variables

· m equations

· homogeneous $\rightarrow$ because of the zeros.

$$\overset{m \times n}{\downarrow} \hat{A} \underset{\hookrightarrow n \times 1}{x} = \underset{\hookrightarrow m \times 1}{0}$$

$$\{x \in \mathbb{F}^n : \hat{A}x = 0\}$$

$\hookrightarrow$ subspace of $\mathbb{F}^n$.

Proposition. Let $A \in \mathcal{M}_{m \times n}(\mathbb{F})$.

The set $\mathcal{N} = \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$ is a subspace of $\mathbb{F}^n$.

Theorem. (Conditions for Subspace)

Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** $\mathbf{0}$ belongs to $\mathcal{U}$.
- **(Closure under Vector Addition).**

For all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, we have $\mathbf{u} \oplus \mathbf{v} \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\mathbf{u} \in \mathcal{U}$, we have $\lambda \otimes \mathbf{u} \in \mathcal{U}$.

$\Rightarrow \mathbf{0} \in \mathcal{N}$ because $\hat{A} \cdot \mathbf{0} = \mathbf{0}$

$\Rightarrow$ let $\mathbf{x}, \mathbf{y} \in \mathcal{N}$ i.e. $\hat{A}\mathbf{x} = \mathbf{0}$ and $\hat{A}\mathbf{y} = \mathbf{0}$

$$\hat{A}(\mathbf{x} + \mathbf{y}) = \hat{A}\mathbf{x} + \hat{A}\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

$$\therefore \mathbf{x} + \mathbf{y} \in \mathcal{N}.$$

$\Rightarrow$ let $\lambda \in \mathbb{F}$, $\mathbf{x} \in \mathcal{N} \Rightarrow \hat{A}\mathbf{x} = \mathbf{0}$

$$\hat{A}(\lambda \mathbf{x}) = \lambda(\hat{A}\mathbf{x}) = \lambda(\mathbf{0}) = \mathbf{0}$$

$$\therefore \lambda \mathbf{x} \in \mathcal{N}$$

$\therefore \mathcal{N}$ is a subspace

MH1201 version

Theorem. (Conditions for Subspace)

Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** $\mathbf{0}$ belongs to $\mathcal{U}$. *
- **(Closure under Vector Addition).**

For all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, we have $\mathbf{u} \oplus \mathbf{v} \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\mathbf{u} \in \mathcal{U}$, we have $\lambda \otimes \mathbf{u} \in \mathcal{U}$.

MHXXXX version

Theorem X. (Conditions for Subspace)

Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Nonempty).** $\mathcal{U}$ is nonempty. *
- **(Closure under Vector Addition).**

For all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, we have $\mathbf{u} \oplus \mathbf{v} \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\mathbf{u} \in \mathcal{U}$, we have $\lambda \otimes \mathbf{u} \in \mathcal{U}$.

Both theorems are correct
→ they imply one another

Proposition. Midterm Qn from last year

Let $\mathcal{U}$ be subset of $\mathcal{V}$ that is closed under vector addition and scalar multiplication.

$\mathbf{0}$ belongs to $\mathcal{U}$ if and only if $\mathcal{U}$ is nonempty.

WTS: $\mathbf{0}$ belongs to $\mathcal{U} \Rightarrow \mathcal{U}$ is non-empty

because $\mathcal{U}$ contains the zero vector

$\mathcal{U}$ is non-empty

$\Rightarrow \mathbf{u}$ belongs to $\mathcal{U}$

$\Rightarrow \mathbf{0} \otimes \mathbf{u}$ belongs to $\mathcal{U}$ because it's closed under $\otimes$

$\Rightarrow$ Since $\mathbf{0} \otimes \mathbf{u} = \mathbf{0}$, we have $\mathbf{0}$ belongs to $\mathcal{U}$

How to Describe a Vector Space?

	Vector space V	Subspace U
MH1200 LA1	$V = \mathbb{R}^n$.	$U \subseteq V$. check U is a subspace $\leftrightarrow$ 3 "axioms"
Middle ground.	$V = \mathbb{R}^n$ $V = M_{m \times n}$ $V = \mathcal{F}(X, \mathbb{R})$ [no need to check ten axioms]	$U \subseteq V$. U is a subspace $\leftrightarrow$ check 3 "conditions"
MH1201 LA2	$V =$ "anything" $\oplus, \otimes =$ "anything". -Ten axioms to check.	$U \subseteq V$. check U is a subspace $\leftrightarrow$ 3 "conditions"

Linear Combinations, Span
Linear Independence
Basis

How to Describe a Vector Space?

- List all vectors. $\rightarrow$ Bad idea

$\rightarrow$ too many vectors

$\rightarrow$ a vector space over $\mathbb{R}$ has infinite vectors (if $|\mathbb{R}| \geq 2$)

- Start with a **finite** set of vectors and provide rules to combine them.

$\rightarrow$ How to "combine" vectors?

$\rightarrow$ How small can this finite set be?

vector addition $(+)$
scalar multiplication $(\times)$ } linear combinations

Linear Combination

Let $S = \{v_1, v_2, \dots, v_n\}$.

- **(Linear Combination).** u is a linear combination of vectors in S if we can write

$$u = (\lambda_1 \otimes v_1) \oplus (\lambda_2 \otimes v_2) \oplus \dots \oplus (\lambda_n \otimes v_n) \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}.$$

Hence forth, we will drop $\oplus$ & $\otimes$

$$u \approx \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n$$

Linear Combination and Span

Let $S = \{v_1, v_2, \dots, v_n\}$.

- **(Linear Combination).** u is a linear combination of vectors in S if we can write

$$u = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}.$$

- **(Span).** The span of S is the set of vectors

$$\text{span}(S) = \{\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n : \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}\}.$$

Linear combination refers to ONE vector

Span refers to a SET of vectors.

Example. Consider $S = \{(2, 1, -3), (1, -2, 4)\} \subseteq \mathbb{R}^3$.

- Is $(17, -4, 2)$ a linear combination of vectors in S ? **Yes**
- Is $(17, -4, 5)$ a linear combination of vectors in S ? **No**

Solve linear equations:

Suppose that

$$(17, -4, 2) = \lambda_1 (2, 1, -3) + \lambda_2 (1, -2, 4)$$

$$\left. \begin{aligned} 17 &= 2\lambda_1 + \lambda_2 \\ -4 &= 2\lambda_1 - 2\lambda_2 \\ 2 &= -3\lambda_1 + 4\lambda_2 \end{aligned} \right\} \begin{aligned} \lambda_1 &= 6 \\ \lambda_2 &= 5 \end{aligned}$$

Suppose that

$$(17, -4, 5) = \lambda_1 (2, 1, -3) + \lambda_2 (1, -2, 4)$$

$$\left. \begin{aligned} 17 &= 2\lambda_1 + \lambda_2 \\ -4 &= 2\lambda_1 - 2\lambda_2 \\ 5 &= -3\lambda_1 + 4\lambda_2 \end{aligned} \right\} \text{No solutions}$$

How to solve?
→ Revisit Cramer's Rule

Example. Consider $S = \{(2, 1, -3), (1, -2, 4)\} \subseteq \mathbb{R}^3$.

- Does $(17, -4, 2)$ belong to $\text{span}(S)$? *Yes*
- Does $(17, -4, 5)$ belong to $\text{span}(S)$? *No*

* $\underline{y}$ is a linear combination of vectors in S iff $\underline{y}$ belongs to $\text{span}(S)$.

Example. Consider $S = \left\{ 1 - x^2, \frac{1}{2}(-x + x^2), \frac{1}{2}(x + x^2) \right\} \subseteq \mathcal{P}_2(\mathbb{R})$.

- Is 1 a linear combination of vectors in S ? *yes*
 - Is x a linear combination of vectors in S ?
 - Is x^2 a linear combination of vectors in S ?
- } yes. $\Rightarrow$ He didn't go thru, he say just trust him.*

$$\begin{aligned}
 1 &= \lambda_1 (1 - x^2) \\
 &+ \lambda_2 \left(-\frac{1}{2}x + \frac{1}{2}x^2 \right) \\
 &+ \lambda_3 \left(\frac{1}{2}x + \frac{1}{2}x^2 \right)
 \end{aligned}$$

coef. of x^0
 $1 = \lambda_1$

coef. of x
 $0 = -\frac{\lambda_2}{2} + \frac{\lambda_3}{2}$

coef. of x^2
 $0 = -\lambda_1 + \frac{\lambda_2}{2} + \frac{\lambda_3}{2}$

Solving this system,
 we have
 $\lambda_1 = \lambda_2 = \lambda_3 = 1$

Example. Let

$$p_1(x) = 1 - x^2,$$

$$p_2(x) = \frac{1}{2}(-x + x^2),$$

$$p_3(x) = \frac{1}{2}(x + x^2), \text{ and}$$

$$S = \{p_1(x), p_2(x), p_3(x)\} \subseteq \mathcal{P}_2(\mathbb{R}).$$

- $1 = p_1(x) + p_2(x) + p_3(x)$
- $x = -p_2(x) + p_3(x)$
- $x^2 = p_2(x) + p_3(x)$

$\therefore 1, x, x^2$ belongs to $\text{span}(S)$

Pick any polynomial in $\mathcal{P}_2(\mathbb{R})$

$$a + bx + cx^2$$

Does $a + bx + cx^2$ belong to $\text{span}(S)$.

$$\left. \begin{array}{l} a = a p_1(x) + a p_2(x) + a p_3(x) \\ bx = -b p_2(x) + b p_3(x) \\ cx^2 = c p_2(x) + c p_3(x) \end{array} \right\} \text{all got linear combination.}$$

$$a + bx + cx^2 = a p_1(x) + (a - b + c) p_2(x) + (a + b + c) p_3(x)$$

Refer to lecture slide. Let S be as defined in the slide.

Let $a + bx + cx^2$ be a polynomial in $\mathcal{P}_2(\mathbb{R})$.

Does $a + bx + cx^2$ belong to $\text{span}(S)$?

Waiting for next clap

Yes

No

Example. Let

$$p_1(x) = 1 - x^2,$$

$$p_2(x) = \frac{1}{2}(-x + x^2),$$

$$p_3(x) = \frac{1}{2}(x + x^2), \text{ and}$$

$$S = \{p_1(x), p_2(x), p_3(x)\} \subseteq \mathcal{P}_2(\mathbb{R}).$$

- $1 = p_1(x) + p_2(x) + p_3(x)$
- $x = -p_2(x) + p_3(x)$
- $x^2 = p_2(x) + p_3(x)$

$$a + bx + cx^2 \in \text{span}(S)$$

$$\Rightarrow \mathcal{P}_2(x) \subseteq \text{span}(S)$$

$$Qn: \text{Is } \mathcal{P}_2(\mathbb{R}) = \text{span}(S) \Rightarrow \text{Yes}$$

$$\begin{array}{c} \cdot x^3 \\ \boxed{\begin{array}{ccc} & & \mathcal{P}_2(\mathbb{R}) \\ p_1(x) & p_2(x) & p_3(x) \end{array}} \end{array}$$

- Is it possible to generate x^3 ? $\rightarrow$ No
- Everything we generate must belong to $\mathcal{P}_2(\mathbb{R})$

Proposition. Let $S = \{v_1, v_2, \dots, v_n\} \subseteq \mathcal{V}$.

(a) $\text{span}(S)$ is a vector subspace of $\mathcal{V}$.

(b) If $\mathcal{U}$ is a vector subspace that contains S , then $\text{span}(S) \subseteq \mathcal{U}$.

a) WTS : $\underline{0} \in \text{span}(S)$

$$\lambda_1 \underline{v}_1 + \lambda_2 \underline{v}_2 + \dots + \lambda_n \underline{v}_n = \underline{0}$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = 0, \dots, \lambda_n = 0$$

Proposition. Let $S = \{v_1, v_2, \dots, v_n\} \subseteq \mathcal{V}$.

(a) $\text{span}(S)$ is a vector subspace of $\mathcal{V}$.

(b) If $\mathcal{U}$ is a vector subspace that contains S , then $\text{span}(S) \subseteq \mathcal{U}$.

a) WTS: closed under $\oplus$

$$u = \lambda_1 v_1 + \dots + \lambda_n v_n \in \text{span}(S)$$

$$v = \mu_1 v_1 + \dots + \mu_n v_n \in \text{span}(S)$$

$$\begin{aligned} u + v &= \lambda_1 v_1 + \mu_1 v_1 + \dots + \lambda_n v_n + \mu_n v_n \\ u + v &= (\lambda_1 + \mu_1) v_1 + \dots + (\lambda_n + \mu_n) v_n \end{aligned} \quad \bigg\} \text{Distributive Axiom}$$

$$\Rightarrow u + v \in \text{span}(S)$$

Proposition. Let $S = \{v_1, v_2, \dots, v_n\} \subseteq \mathcal{V}$.

(a) $\text{span}(S)$ is a vector subspace of $\mathcal{V}$.

(b) If $\mathcal{U}$ is a vector subspace that contains S , then $\text{span}(S) \subseteq \mathcal{U}$.

a) WTS: closed under $\otimes$

Let $\lambda \in \mathbb{F}$, $u \in \text{span}(S)$

$$u = \lambda_1 v_1 + \dots + \lambda_n v_n$$

$$\begin{aligned} \Rightarrow \lambda u &= \lambda(\lambda_1 v_1 + \dots + \lambda_n v_n) \\ &= (\lambda \lambda_1) v_1 + \dots + (\lambda \lambda_n) v_n \end{aligned}$$

$$\Downarrow$$
$$\lambda u \in \text{span}(S)$$

$\therefore \text{span}(S)$ is subspace

Proposition. Let $S = \{v_1, v_2, \dots, v_n\} \subseteq \mathcal{V}$.

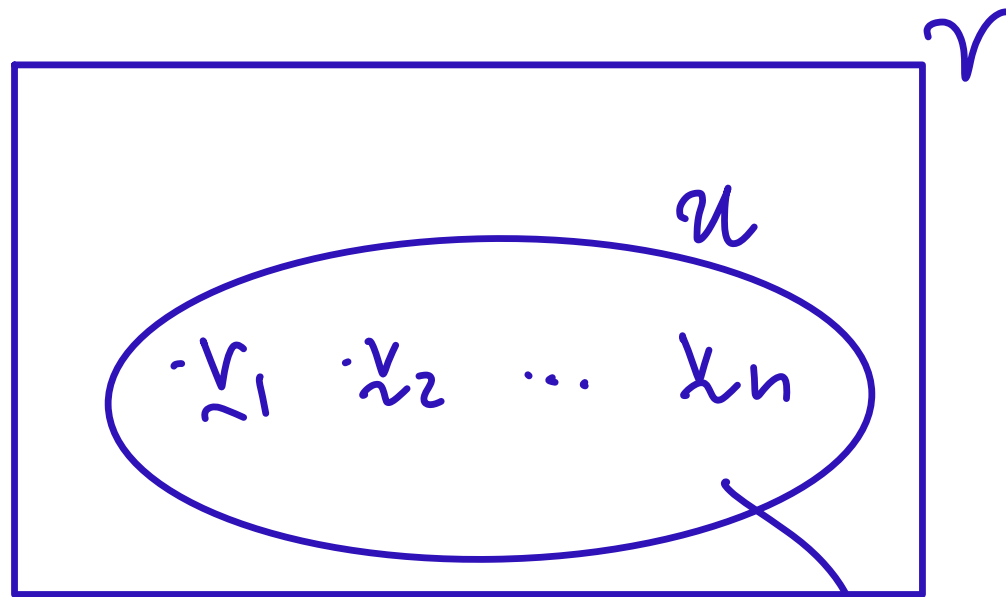
(a) $\text{span}(S)$ is a vector subspace of $\mathcal{V}$.

(b) If $\mathcal{U}$ is a vector subspace that contains S , then $\text{span}(S) \subseteq \mathcal{U}$.

b)

"Picture"

* Proof in Colab
→ not expected to
be able to prove it



→ $\text{span}(S)$ cannot be bigger
than $\mathcal{U}$.

Using SymPy

Let's solve

$$2\lambda_1 + \lambda_2 = 17$$

$$\lambda_1 - 2\lambda_2 = -4$$

$$-3\lambda_1 + 4\lambda_2 = 2$$

$$\Rightarrow \lambda_1 = 6, \lambda_2 = 5$$

For this course, if have
system of linear eqn don't
need show working to
get to
REF
↓
can just
write
final ans

```
1  # NOTE: we use lamb to represent lambda
2
3  import sympy as sympy
4  lamb1 = sympy.var('lamb1')
5  lamb2 = sympy.var('lamb2')
6
7  eq1 = 2*lamb1 + lamb2 - (17)
8  eq2 = 1*lamb1 - 2*lamb2 - (-4)
9  eq3 = -3*lamb1 + 4*lamb2 - (2)
10
11 sol = sympy.solve([eq1,eq2,eq3], dict=True)
12 if len(sol) > 0:
13     print("Solution : ", sol)
14 else:
15     print(sol)
16     print("NO SOLUTION EXIST!")
```